

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
%matplotlib inline
warnings.filterwarnings('ignore')
pd.options.display.max_columns=99
```

```
df=pd.read_csv("/content/Online_Retail.csv", encoding='latin1')
```

```
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/10 8:26	2.55	17850.0	United Kingdom	
1	536365	71053	WHITE METAL LANTERN	6	12/1/10 8:26	3.39	17850.0	United Kingdom	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/10 8:26	2.75	17850.0	United Kingdom	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/10 8:26	3.39	17850.0	United Kingdom	
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/10 8:26	3.39	17850.0	United Kingdom	

```
df.shape
```

```
(453489, 8)
```

```
df.tail()
```

```
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
453484	575482	22224	WHITE LOVEBIRD LANTERN	4	11/9/11 16:33	2.95	15916.0	United Kingdom
453485	575482	23355	HOT WATER BOTTLE KEEP CALM	2	11/9/11 16:33	4.95	15916.0	United Kingdom
453486	575482	22114	HOT WATER BOTTLE TEA AND SYMPATHY	2	11/9/11 16:33	4.25	15916.0	United Kingdom
453487	575482	84030E	ENGLISH ROSE HOT WATER BOTTLE	2	11/9/11 16:33	4.25	15916.0	United Kingdom
453488	575482	22639	SET OF 4 NAPKIN CHARMS HEARTS	4	11/9/11 16:33	2.55	NaN	NaN

```
df.describe()
```

	Quantity	UnitPrice	CustomerID
count	453489.000000	453489.000000	341144.000000
mean	9.775620	4.686967	15282.592530
std	165.444067	100.142865	1712.270347
min	-74215.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13911.000000
50%	3.000000	2.080000	15156.000000
75%	11.000000	4.130000	16779.000000
max	74215.000000	38970.000000	18287.000000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 453489 entries, 0 to 453488
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -

```

```
0 InvoiceNo 453489 non-null object
1 StockCode 453489 non-null object
2 Description 452115 non-null object
3 Quantity 453489 non-null int64
4 InvoiceDate 453489 non-null object
5 UnitPrice 453489 non-null float64
6 CustomerID 341144 non-null float64
7 Country 453488 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 27.7+ MB
```

```
df.isnull().sum()
```

	0
InvoiceNo	0
StockCode	0
Description	1374
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	112345
Country	1

dtype: int64

```
plt.style.use("fivethirtyeight")
```

```
x_question=df.iloc[:,6:33]
q_mean=x_question.mean(axis=0, numeric_only=True)
total_mean=q_mean.mean()
```

```
q_mean = q_mean.reset_index(level=0)
q_mean = q_mean.rename(columns={0: 'Mean Value'})
q_mean.head()
```

level_0	index	Mean Value	
0	0 CustomerID	15282.59253	

```
# q_mean=q_mean.to_frame('mean') # This line is not needed as q_mean is already a DataFrame
```

```
plt.figure(figsize=(14,7))
sns.barplot(x='index',y='Mean Value',data=q_mean)
```

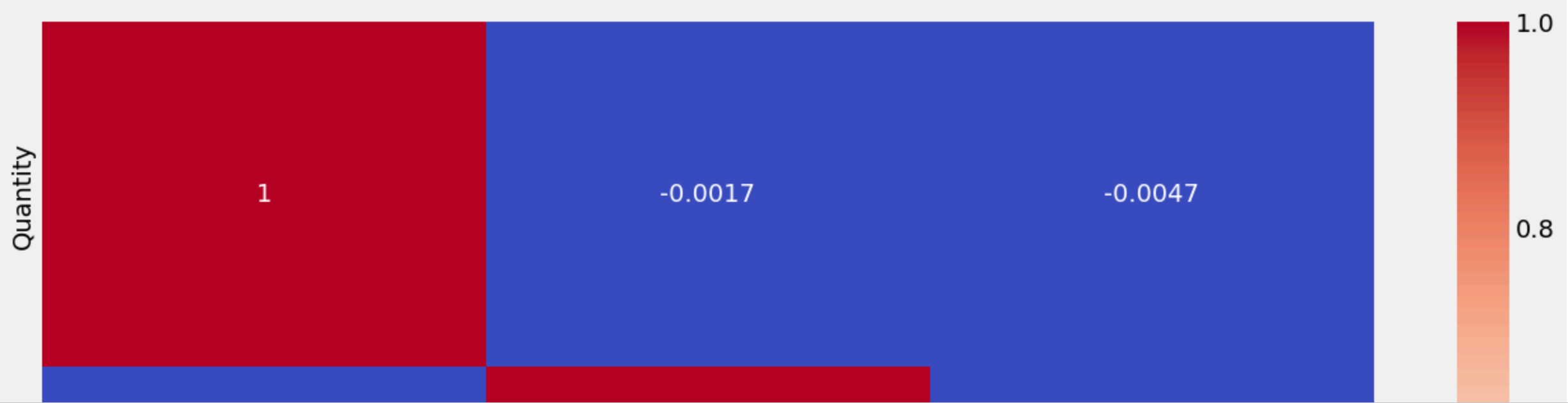
<Axes: xlabel='index', ylabel='Mean Value'>



```
corr=df.corr(numeric_only=True)
```

```
plt.figure(figsize=(15,10))
sns.heatmap(corr,annot=True,cmap="coolwarm")#blue
```

<Axes: >



```
x=df.iloc[:,6:7]
```

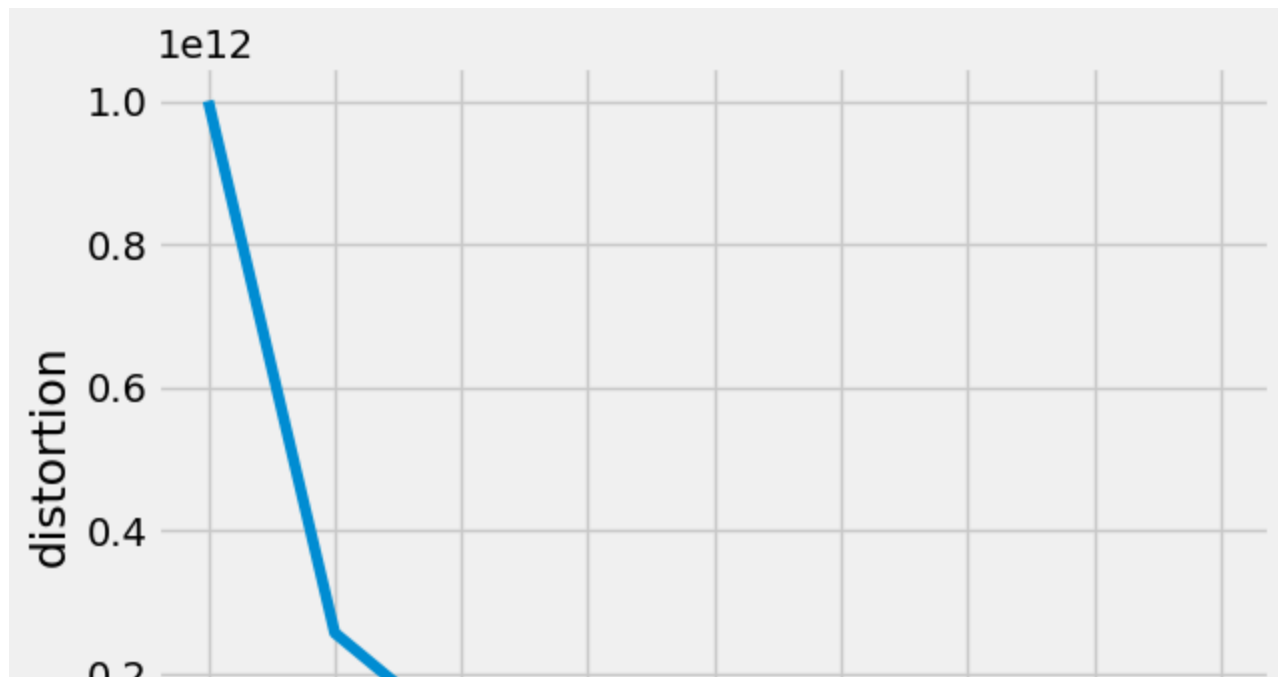
```
from sklearn.decomposition import PCA
PCA=PCA(n_components=1,random_state=42)
x = x.dropna()
x_PCA=PCA.fit_transform(x)
```

```
x_PCA
```

```
array([[2567.40747016],  
       [2567.40747016],  
       [2567.40747016],  
       ...,  
       [ 633.40747016],  
       [ 633.40747016],  
       [ 633.40747016]])
```

```
from sklearn.cluster import KMeans  
distortions = []  
cluster_range=range(1,10)  
for i in cluster_range:  
    km=KMeans(n_clusters=i,random_state=42,n_init=10)  
    km.fit(x_PCA)  
    distortions.append(km.inertia_)
```

```
plt.plot(cluster_range,distortions)  
plt.xlabel("Number of Clusters")  
plt.ylabel("distortion")  
plt.show()
```



```
model=KMeans(n_clusters=3,random_state=42,n_init=10)  
model.fit(x_PCA)  
y=model.predict(x_PCA)
```

```
plt.scatter(x_PCA[y==0, 0], y[y==0], s=50, c='red', label='Cluster 0')  
plt.scatter(x_PCA[y==1, 0], y[y==1], s=50, c='blue', label='Cluster 1')  
plt.scatter(x_PCA[y==2, 0], y[y==2], s=50, c='green', label='Cluster 2')  
plt.scatter(model.cluster_centers_[0, 0], model.cluster_centers_[0, 1], s=50, c='red', label='Cluster 0 Center')  
plt.scatter(model.cluster_centers_[1, 0], model.cluster_centers_[1, 1], s=50, c='blue', label='Cluster 1 Center')  
plt.scatter(model.cluster_centers_[2, 0], model.cluster_centers_[2, 1], s=50, c='green', label='Cluster 2 Center')  
plt.legend()
```

```
plt.title('Clusters of customers')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```

